

AA 2025/2026

# Machine Learning for Modelling: *Supervised Learning*

Simone Bianco

1

AA 2025/2026

# CNN architectures

61

## Applications

- 14,197,122 images
- 21,841 synsets

# IMAGENET



J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li and L. Fei-Fei, ImageNet: A Large-Scale Hierarchical Image Database, *IEEE Computer Vision and Pattern Recognition (CVPR)*, 2009.

62

## Applications

- IMAGENET
- Image classification

### Large Scale Visual Recognition Challenge

The Image Classification Challenge:  
1,000 object classes  
1,431,167 images

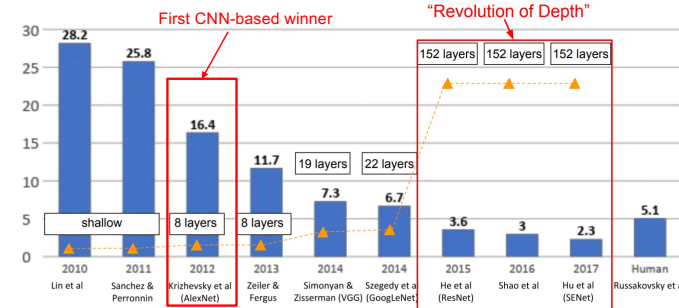
|                                                                                       |                                                                             |                                                                                       |                                                                              |                                                                                       |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  | <b>Output:</b><br>Scale<br>T-shirt<br>Steel drum<br>Drumstick<br>Mud turtle |  | <b>Output:</b><br>Scale<br>T-shirt<br>Giant panda<br>Drumstick<br>Mud turtle |  |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|

63

## CNN Architectures

### Imagenet classification error

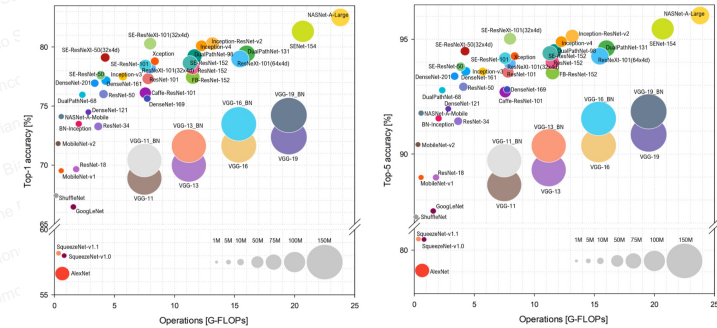
ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners



64

## CNN Architectures

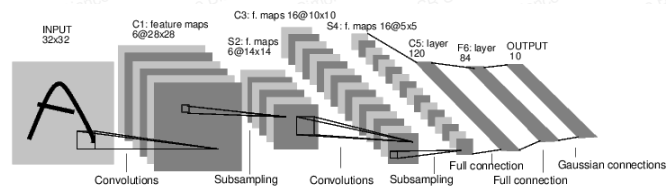
### Imagenet classification error



65

## CNN Architectures

### LeNet (LeNet-5)

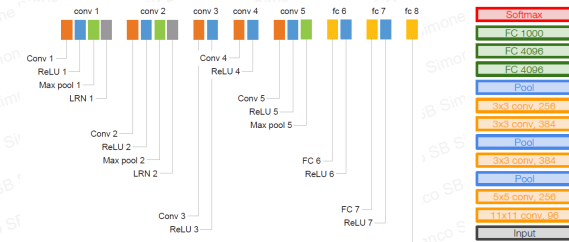


LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278-2324.

66

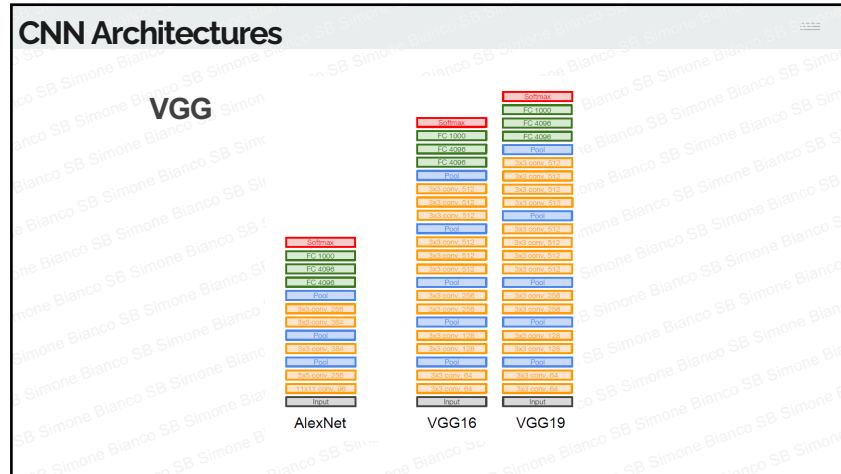
## CNN Architectures

### AlexNet

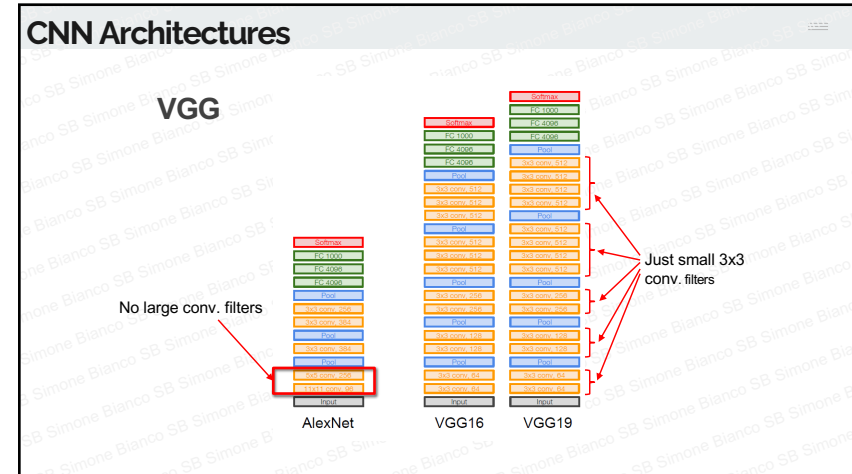


Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems* (pp. 1097-1105).

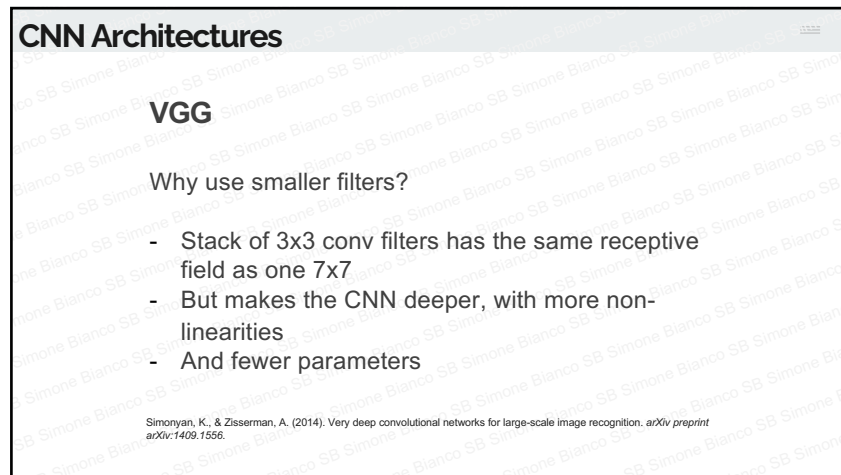
67



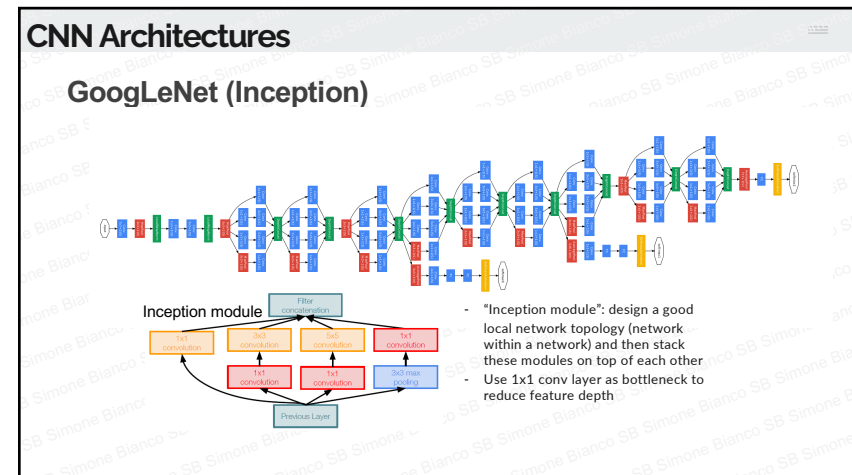
68



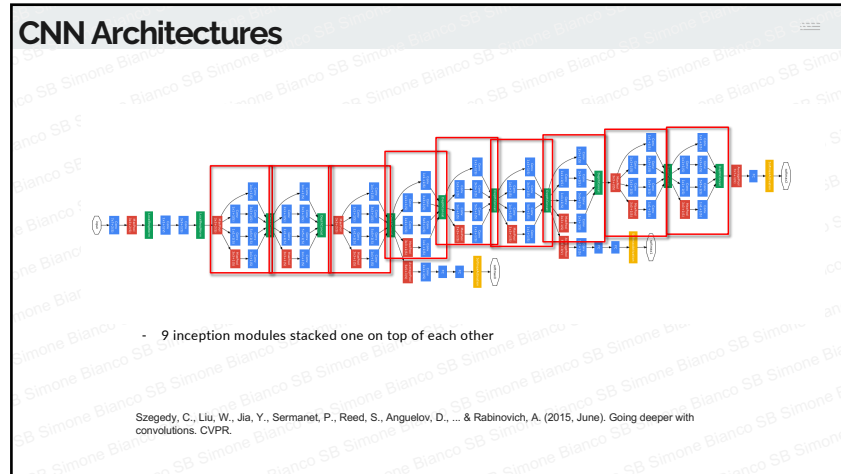
69



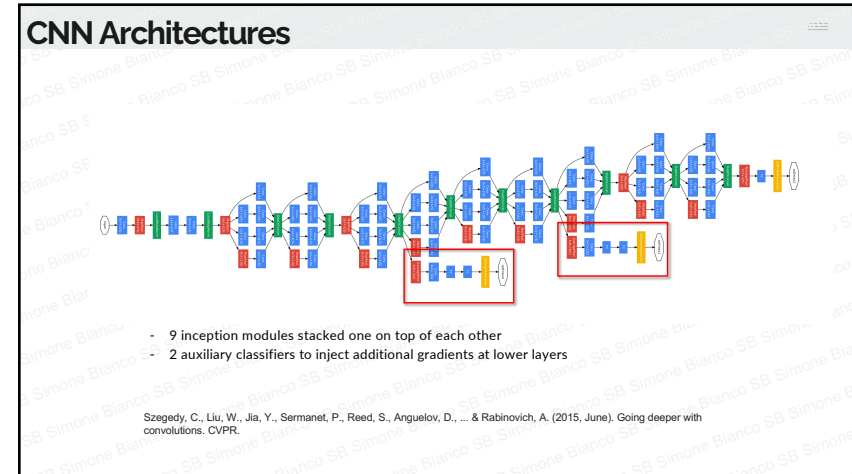
70



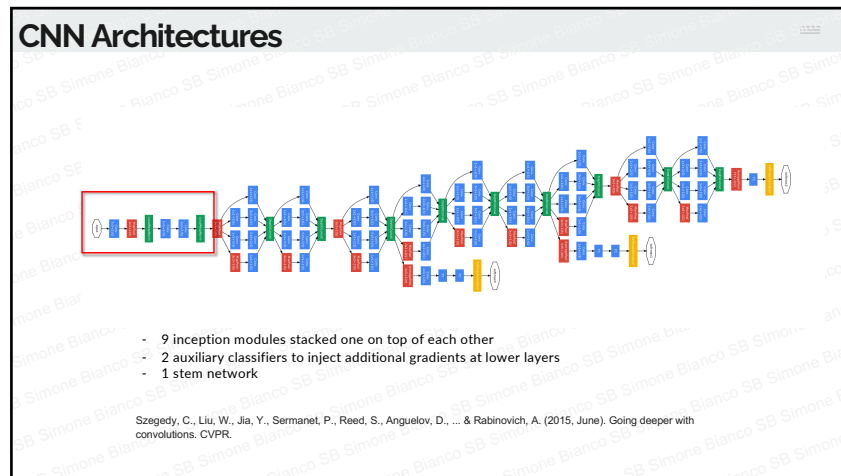
71



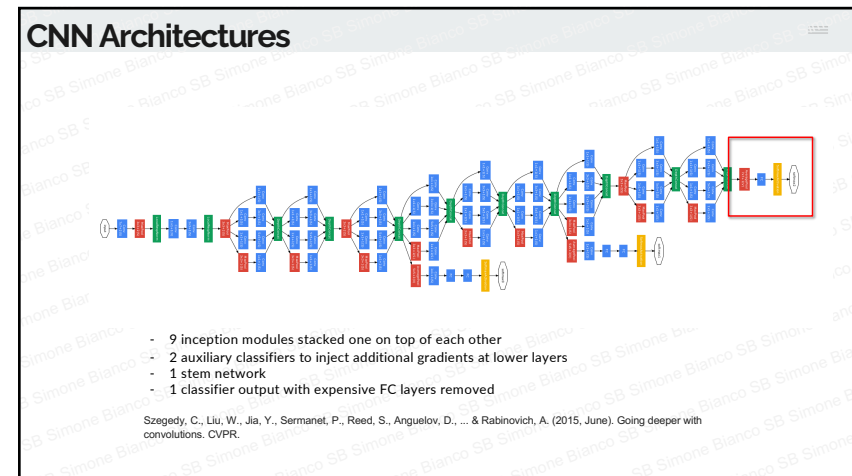
72



73



74

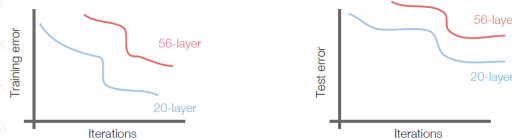


75

## CNN Architectures

### ResNet

- So far, the deeper the better
- Can we continue on adding layer and go deeper?



He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

76

## CNN Architectures

### ResNet

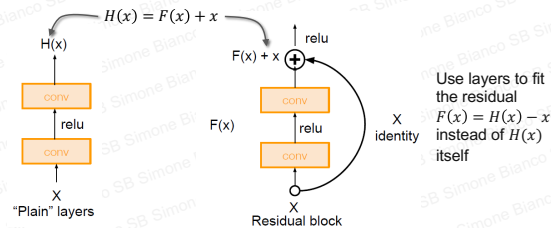
- So far, the deeper the better
- Can we continue on adding layer and go deeper?
- Unfortunately no
- Hypothesis: the problem is an optimization problem, deeper models are harder to optimize (e.g. vanishing gradients)

77

## CNN Architectures

### ResNet

Solution: Use network layers to fit a residual mapping instead of directly trying to fit a desired underlying mapping

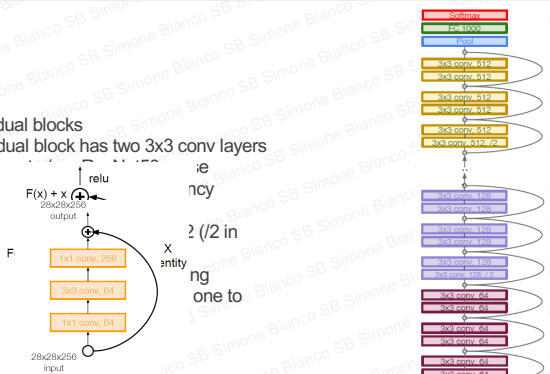


78

## CNN Architectures

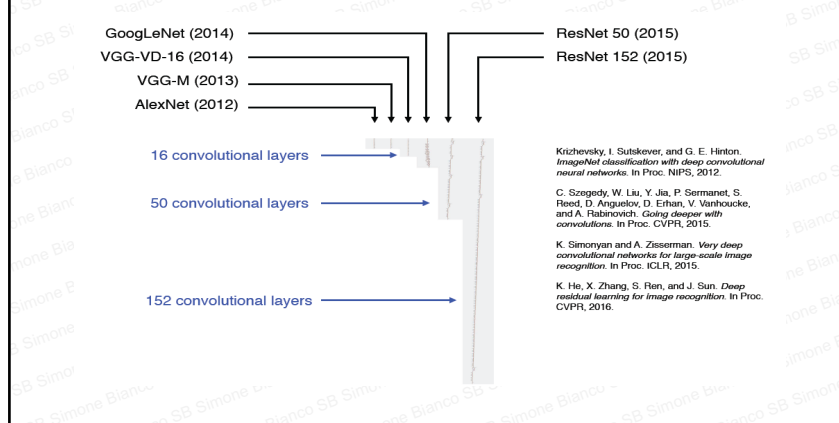
### ResNet

- Stack residual blocks
- Every residual block has two 3x3 conv layers
- For deeper bottleneck
- Periodically downsample each time
- Additional No FC layer map to the



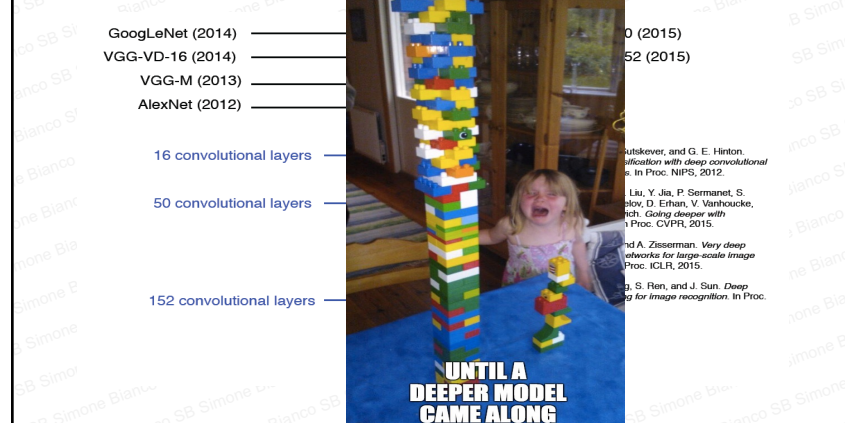
79

## How deep is enough?



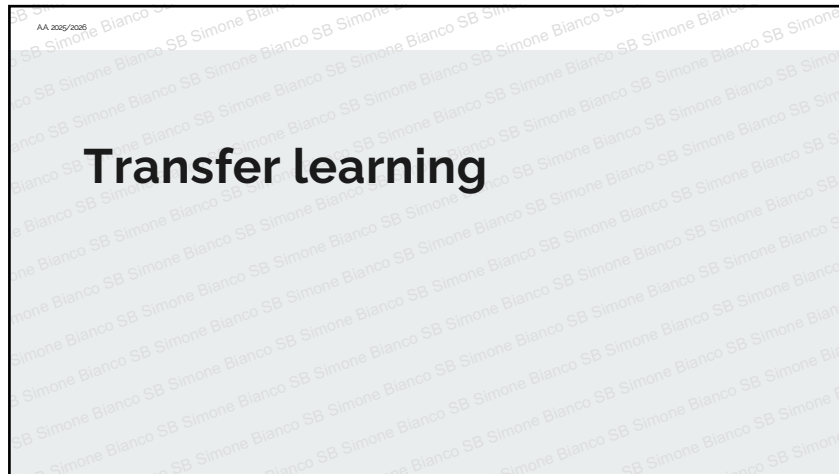
80

## How deep is enough?



81

## Transfer learning



82

## Transfer learning

- All the best performing architectures have millions of parameters to learn
- In order to tune them, very large databases are needed (e.g., rule of 10: 10x samples wrt number of parameters to train)



- What if we do not have such amount of data? Can we still use deep learning?

83



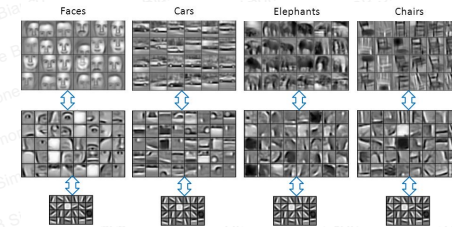
## Transfer learning

- In practice, very few people train an entire CNN from scratch, because it is relatively rare to have a dataset of sufficient size.
- It is common to pretrain a CNN on a very large dataset (e.g., ImageNet which contains 1.2 million images with 1000 categories)
- and then use the CNN either for fine-tuning or as a fixed feature extractor for the task of interest.

84

## Transfer learning

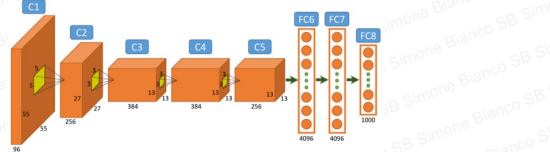
- Why does it work?
- Motivated by the observation that the earlier features of a CNN contain more generic features (e.g., edge detectors or color blob detectors) that should be useful to many tasks
- Later layers of the CNN becomes progressively more specific to the details of the classes contained in the original dataset.



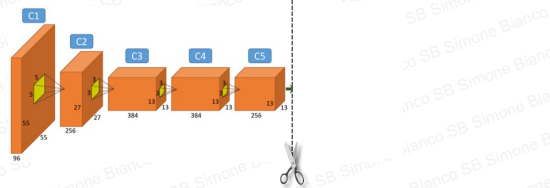
85

## CNN fine-tuning

- Task 1 (e.g. ImageNet, 1000 categories)



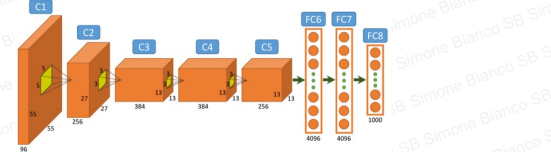
- Task 2 (10 categories)



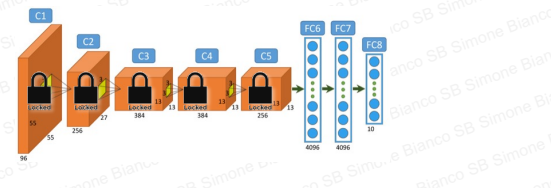
86

## CNN fine-tuning

- Task 1 (e.g. ImageNet, 1000 categories)

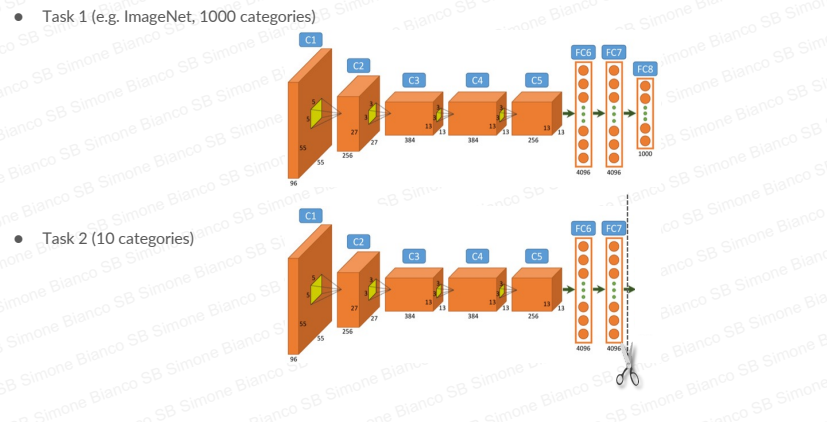


- Task 2 (10 categories)



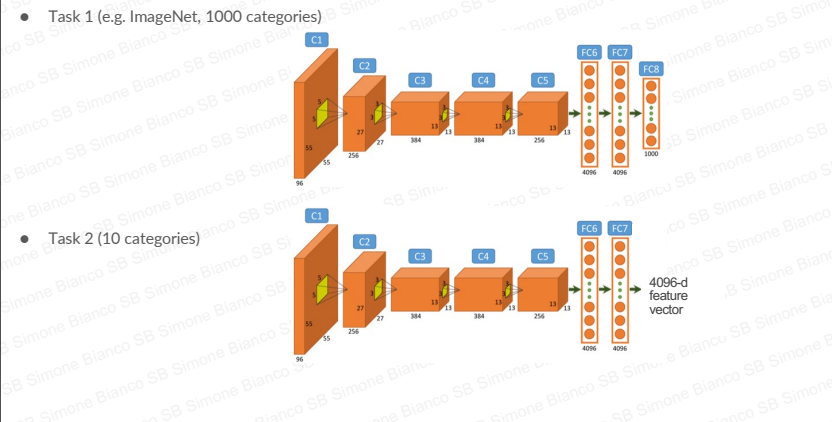
87

## CNN as feature extractor



88

## CNN as feature extractor



89

## How to decide what type of TL you should perform?

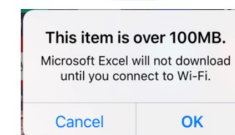
- New dataset is **small and similar** to original dataset:
  - Train a linear classifier on CNN features from higher layers
- New dataset is **large and similar** to original dataset
  - Fine-tune the CNN
- New dataset is **small but very different** from original dataset
  - Train a linear classifier on CNN features from lower layers
- New dataset is **large and very different** from original dataset
  - Train CNN from scratch or fine-tune it

90

## Model Compression

The problem: If running DNN on Mobile...

APP developers suffer from the model size!



91

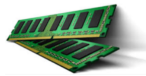
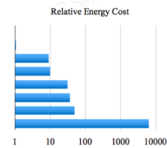


## Model Compression

The problem: If running DNN on Mobile...

HW engineers suffer from the model size! (embedded systems, limited resources)

| Operation            | Energy [pJ] | Relative Cost |
|----------------------|-------------|---------------|
| 32 bit int ADD       | 0.1         | 1             |
| 32 bit float ADD     | 0.9         | 9             |
| 32 bit Register File | 1           | 10            |
| 32 bit int MULT      | 3.1         | 31            |
| 32 bit float MULT    | 3.7         | 37            |
| 32 bit SRAM Cache    | 5           | 50            |
| 32 bit DRAM Memory   | 640         | 6400          |



92

## Model Compression

The problem: If running DNN on the Cloud...

Network  
Delay

Power  
Budget

User  
Privacy

Intelligent but Inefficient

93

## Model Compression

**Smaller Size**  
Compress Mobile App  
Size by 35x-50x

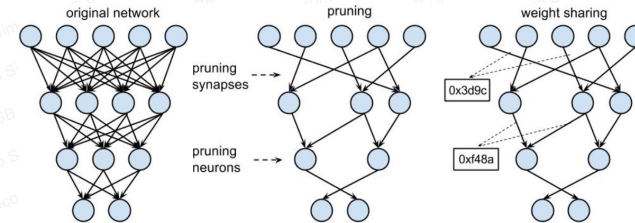
**Accuracy**  
no loss of accuracy  
improved accuracy

**Speedup**  
make inference faster

- AlexNet: 35x, 240MB => 6.9MB
- VGG16: 49x 552MB => 11.3MB
- Both with no loss of accuracy on ImageNet
- Weights fits on-chip SRAM, taking 120x less energy than DRAM

94

## Model Compression



95

## Model Compression

- 50 trillions of synapses are generated in the human brain during the first few months of birth.
- 1 year old, peaked at 1000 trillions
- Pruning begins to occur.
- 10 years old, a child has nearly 500 trillions synapses
- This 'pruning' mechanism removes redundant connections in the brain.

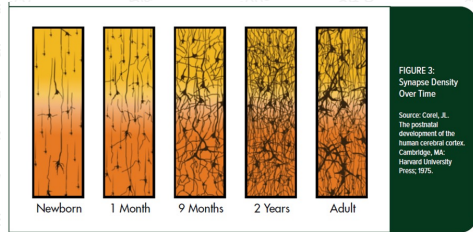


FIGURE 3:  
Synapse Density  
Over Time

Source: Corel, R.  
The postnatal  
development of the  
human cerebral cortex.  
Cambridge, MA:  
Harvard University  
Press, 1975.